ROLLUPOS APPLICATION USER MANUAL • FIELD & PRODUCTIVITY

# 🧭 CSC Site Discovery

Offline-First HVAC/BAS Field Survey Companion • Division 23 Checklists, CLAHE Nameplate OCR & Building Mapper

Version: v1.0.1 | Build Hash: #aeb18853 | 🔒 Per-User Authorized Cartridge | Auto-Synced: 2026-10-24 22:15

## 1. ⚡ 60-Second Quickstart Guide

**1** **Initialize Your Site Walk (`⚙️ Walk / API Vault`)**
Tap `⚙️ Walk / API Vault` (or `Walk / Export`) to enter the Facility Name, CSC Job/Estimate Number (`CSC-26-xxxx`), and optionally save a personal Gemini API key in your private on-device Vault.

**2** **Add Equipment Stations & Select System Type**
Create stations for AHUs, RTUs, Chillers, Boilers, Hydronic Pumps, VAVs, or Main BAS TCPs. The app dynamically loads the matchingCSC Division 23 (`RFI-23-001` through `RFI-23-011`) engineering checklist.

**3** **Capture 3-Slot Burst Photos w/ Stamped-Metal CLAHE**
Snap (1) Wide Equipment Context, (2) Rusted/Stamped Nameplate (auto-enhanced via local CLAHE contrast boosting + serial vintage decoding), and (3) Inside Control Cabinet / Termination Strip.

**4** **Pin Evacuation Floorplan & Export Survey Package**
Snap the wall-mounted Building Evacuation Map, tap to drop equipment pins with GNSS/Compass bearings, record 150Hz high-pass filtered voice walkthroughs, and export the complete JSON/CSV package via Android Share Sheet.

## 2. 🧭 Screen-by-Screen & Navigation Map

| SCREEN / TAB / MODAL | WHAT IT CONTROLS & HOW TO USE IT |
|---|---|
| 📋 **Equipment Station List (Left/Top)** | Add, filter, and switch between rooftop, mechanical room, and terminal unit stations during a walk. |
| 🔍 **Multimodal Station Inspector** | Dynamic RFI-23-001..011 checklists, 3-slot CLAHE camera capture, serial age decoder, and 150Hz filtered voice notes. |
| 🗺️ **4-Layer Building & Evac Mapper** | Pin equipment stations directly onto photographed floorplans/evacuation maps with compass orientation and Laplacian video sharpness scoring. |
| 🧮 **Field Valve / VA / Labor Sizers** | On-site deterministic Cv valve sizer (4–5 psi ΔP / β ≥ 0.50), NEC Art. 725 80% Class 2 panel VA sizer, and rough-in labor estimator. |
| ⚙️ **Session Manager, Vault & Export** | Manage walk metadata, optional Gemini OCR key, Golden Demo Walk loader, and 1-tap JSON/CSV exports. |

## 3. 🔬 In-Depth Feature Guide & Workflows

### CSC Overlay & Safety Checklist Enforcement (`RFI-23-001`..`011`)

Prompts field engineers to verify factory OEM BACnet cards (`Overlay, Never Gut`), hardwired VFD HOA/speed split (`RFI-23-006`), DPDT manual-reset serpentine freeze stats (`RFI-23-007`), Div 23 21 13 inline valve/well boundaries (`RFI-23-005`), and dedicated Div 26 120VAC 15A TCP circuits (`RFI-23-009`).

### CLAHE Stamped-Aluminum Nameplate Enhancer & Offline Queue

Applies local canvas Contrast-Limited Adaptive Histogram Equalization so sun-bleached or rusted outdoor RTU/chiller data plates become legible instantly—working 100% offline in mechanical basements.

### 150Hz Mechanical-Room High-Pass Audio Filter

Uses WebAudio high-pass filtering to strip low-frequency fan/pump rumble (`<150Hz`) from walk-and-talk voice memos for crystal-clear Whisper transcription back at the office.

## 4. ⚙️ Settings, Secret Vault & Multi-User Data Isolation

| SETTING KEY | FIELD DESCRIPTION | TYPE | DEFAULT |
|---|---|---|---|
| `default_job_number` | Default CSC Job / Estimate Number | `text` | `CSC-26-NEW` |
| `engineer_name` | Lead Engineer / Estimator Name | `text` | `CSC Field Engineer` |

🔒 **Storage Namespace Pattern:** `rollup::v1::u::<google-email>::app::csc-site-discovery::*`

📦 **Active App Storage Keys:** `__user_config__` (Managed automatically)

🔐 **Private On-Device Vault Secrets:** None required (uses Hub session)

Isolated Per-User Storage (RollupBridge) · Mobile Camera / Multimodal Capture · WebAudio / Voice Recording Pipeline · Live GNSS / Compass Telemetry · Android Share Sheet & CSV/JSON Export · Encrypted On-Device Secret Vault

## 5. 🎛️ Live UI Controls Reference & Recent Updates

**Interactive Controls Detected in Current Build ( `v1.0.1` ):**

🚨 RFI-23-011 HARD-STOP ACTIVE · 🛰️ Lock GPS/Compass · ⬆️ +1 · ⬇️ -1 · 🌞/🌙 Glare · ⚙️ Walk / Export · 📷 Snap Evacuation / Floor Map · 🗑️ Clear Map · 📤 Export CSV · 🗑️ Delete Pin · ☑️ 1. Checklists · 📸 2. Nameplate & OCR · 🎥 3. Video Mapper · 🎙️ 4. Walk-Talk Mic

**Pro Tips & Best Practices:**

- Always enforce NFPA 70E Lockout/Tagout (LOTO) and PPE before opening live control cabinets for internal termination photos.
- No cell signal in a sub-basement chiller plant? Leave the API key blank—every photo, checklist, and voice clip saves to IndexedDB/RollupBridge offline.

**Recent Cartridge Update Log (Auto-Tracked on Publish):**

- `22d612c` (2026-10-24) — RollupOS Hub & Mini-App sync (2026-10-24 19:59)
- `0c21c6e` (2026-10-24) — RollupOS Hub & Mini-App sync (2026-10-24 16:38)